

# Resilience Experiment Report — CE-408

Ahmed Musharaf · 2026-05-11 · AWS ECS Fargate, us-east-1 · Load: k6 · 20 VUs · baseline.js

## BASELINE

|                                                               |                                                                  |                                                            |                                                               |                                                             |
|---------------------------------------------------------------|------------------------------------------------------------------|------------------------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------------|
| <div>ERROR RATE</div> <div>0%</div> <div>5,960 requests</div> | <div>P95 LATENCY</div> <div>14ms</div> <div>Catalog + Cart</div> | <div>ORDERS P95</div> <div>50ms</div> <div>pre-chaos</div> | <div>THROUGHPUT</div> <div>20/s</div> <div>steady state</div> | <div>TASKS / SVC</div> <div>1</div> <div>all services</div> |
|---------------------------------------------------------------|------------------------------------------------------------------|------------------------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------------|

## CHAOS EXPERIMENTS

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                      |                   |                |       |                |       |                |       |               |        |            |   |              |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |                      |          |       |                  |        |                   |         |                     |       |            |   |                         |    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-------------------|----------------|-------|----------------|-------|----------------|-------|---------------|--------|------------|---|--------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------------------|----------|-------|------------------|--------|-------------------|---------|---------------------|-------|------------|---|-------------------------|----|
| <div>EXPERIMENT 1</div> <div>Task Termination — Catalog</div> <table><tr><td>Method</td><td>aws ecs stop-task</td></tr><tr><td>Fault injected</td><td>17:45</td></tr><tr><td>Task count → 0</td><td>17:45</td></tr><tr><td>Task count → 1</td><td>17:47</td></tr><tr><td>Recovery time</td><td>~2 min</td></tr><tr><td>5xx errors</td><td>0</td></tr><tr><td>Blast radius</td><td>Catalog only</td></tr></table> <div><div>✓ Zero errors</div><div>✓ Auto-recovered</div><div>✓ Fault isolated</div></div> | Method               | aws ecs stop-task | Fault injected | 17:45 | Task count → 0 | 17:45 | Task count → 1 | 17:47 | Recovery time | ~2 min | 5xx errors | 0 | Blast radius | Catalog only | <div>EXPERIMENT 2</div> <div>Latency Injection — Orders +500ms</div> <table><tr><td>Method</td><td>CHAOS_LATENCY_MS=500</td></tr><tr><td>Duration</td><td>180 s</td></tr><tr><td>Baseline latency</td><td>~50 ms</td></tr><tr><td>Chaos avg latency</td><td>~550 ms</td></tr><tr><td>Peak (deploy spike)</td><td>3.6 s</td></tr><tr><td>5xx errors</td><td>0</td></tr><tr><td>Cart / Catalog affected</td><td>No</td></tr></table> <div><div>✓ 11× latency observed</div><div>✓ Auto-rollback</div><div>✓ Fault isolated</div></div> | Method | CHAOS_LATENCY_MS=500 | Duration | 180 s | Baseline latency | ~50 ms | Chaos avg latency | ~550 ms | Peak (deploy spike) | 3.6 s | 5xx errors | 0 | Cart / Catalog affected | No |
| Method                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | aws ecs stop-task    |                   |                |       |                |       |                |       |               |        |            |   |              |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |                      |          |       |                  |        |                   |         |                     |       |            |   |                         |    |
| Fault injected                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 17:45                |                   |                |       |                |       |                |       |               |        |            |   |              |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |                      |          |       |                  |        |                   |         |                     |       |            |   |                         |    |
| Task count → 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 17:45                |                   |                |       |                |       |                |       |               |        |            |   |              |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |                      |          |       |                  |        |                   |         |                     |       |            |   |                         |    |
| Task count → 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 17:47                |                   |                |       |                |       |                |       |               |        |            |   |              |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |                      |          |       |                  |        |                   |         |                     |       |            |   |                         |    |
| Recovery time                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | ~2 min               |                   |                |       |                |       |                |       |               |        |            |   |              |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |                      |          |       |                  |        |                   |         |                     |       |            |   |                         |    |
| 5xx errors                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 0                    |                   |                |       |                |       |                |       |               |        |            |   |              |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |                      |          |       |                  |        |                   |         |                     |       |            |   |                         |    |
| Blast radius                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Catalog only         |                   |                |       |                |       |                |       |               |        |            |   |              |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |                      |          |       |                  |        |                   |         |                     |       |            |   |                         |    |
| Method                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | CHAOS_LATENCY_MS=500 |                   |                |       |                |       |                |       |               |        |            |   |              |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |                      |          |       |                  |        |                   |         |                     |       |            |   |                         |    |
| Duration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 180 s                |                   |                |       |                |       |                |       |               |        |            |   |              |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |                      |          |       |                  |        |                   |         |                     |       |            |   |                         |    |
| Baseline latency                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | ~50 ms               |                   |                |       |                |       |                |       |               |        |            |   |              |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |                      |          |       |                  |        |                   |         |                     |       |            |   |                         |    |
| Chaos avg latency                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | ~550 ms              |                   |                |       |                |       |                |       |               |        |            |   |              |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |                      |          |       |                  |        |                   |         |                     |       |            |   |                         |    |
| Peak (deploy spike)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 3.6 s                |                   |                |       |                |       |                |       |               |        |            |   |              |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |                      |          |       |                  |        |                   |         |                     |       |            |   |                         |    |
| 5xx errors                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 0                    |                   |                |       |                |       |                |       |               |        |            |   |              |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |                      |          |       |                  |        |                   |         |                     |       |            |   |                         |    |
| Cart / Catalog affected                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | No                   |                   |                |       |                |       |                |       |               |        |            |   |              |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |                      |          |       |                  |        |                   |         |                     |       |            |   |                         |    |

GIKI · CE-408 Cloud Computing · ECS Fargate + ALB + RDS PostgreSQL + DynamoDB + SQS